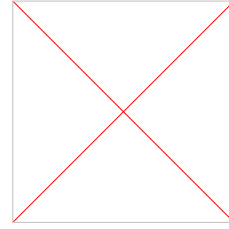




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DI BRESCIA

Experimental characterization of MEMS resonators

DIMIN - Digital twins of nonlinear Microstructures with iNnovative model-order-reduction strategies

M. Zini, A. Nastro, M. Ferrari

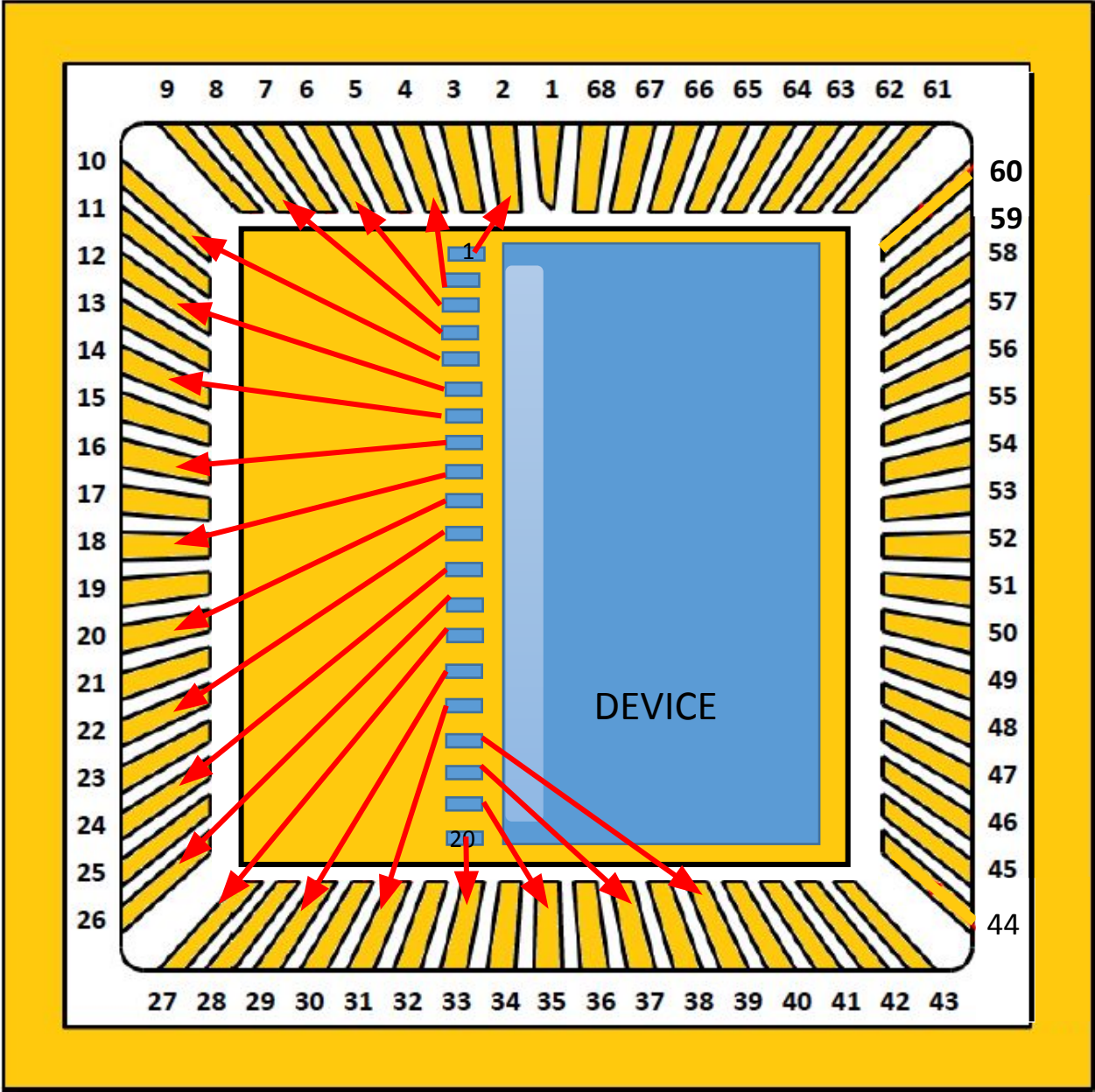
D3.1: Experimental characterization of PoC device and validation of all the procedures implemented for analysis and identification

Dept. Information Engineering

University of Brescia



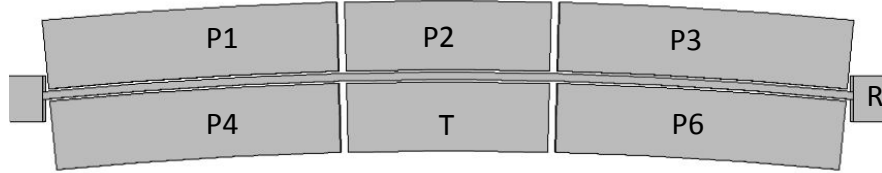
PLCC68 configuration



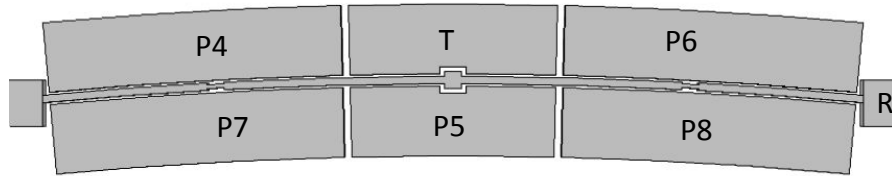
Pad #	Pad	Pin PLCC
1	P6	2
2	P3	4
3	P2	6
4	P1	8
5	P4	10
6	P7	12
7	P5	14
8	P8	16
9	P11	18
10	P10	20
11	P9	22
12	T	24
13	P12	26
14	R	27
15	P13	29
16	P14	31
17	P17	39
18	P16	37
19	P15	35
20	G	33

MEMS resonators

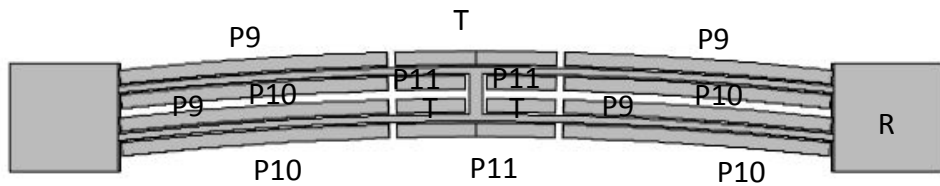
Arco_01



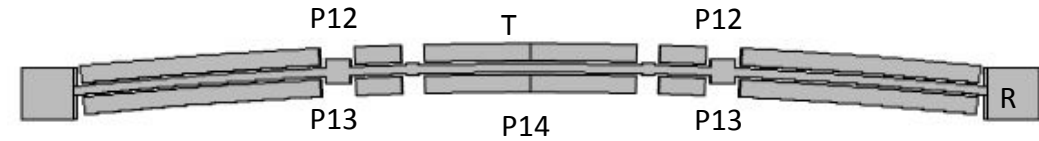
Arco_10b



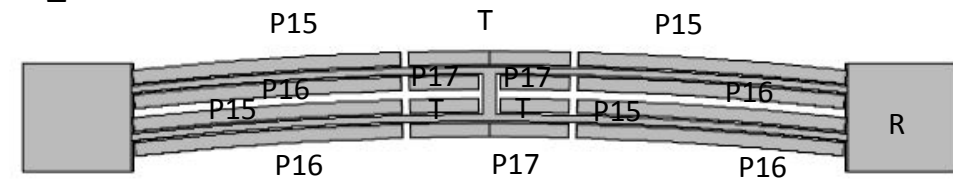
Arco_07



Arco_06



Arco_07b



P6

P3

P2

P1

P4

P7

P5

P8

P11

P10

P9

T

P12

R

P13

P14

P17

P16

P15

G

2

4

6

8

10

12

14

16

18

20

22

24

26

27

29

31

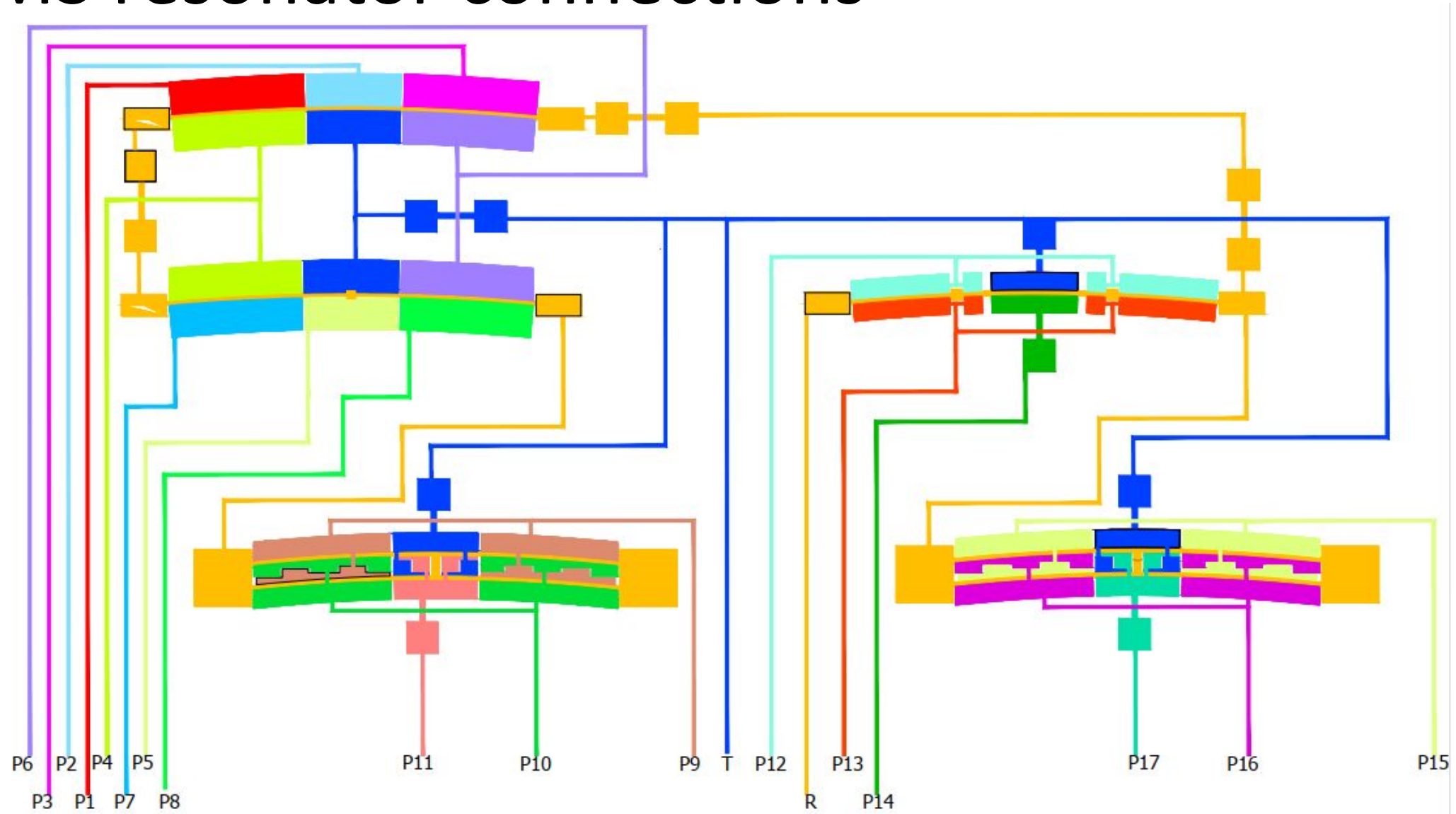
39

37

35

33

MEMS resonator connections



P6

P3

P2

P1

P4

P7

P5

P8

P11

P10

P9

T

P12

R

P13

P14

P17

P16

P15

G

2

4

6

8

10

12

14

16

18

20

22

24

26

27

29

31

39

37

35

33

Quality factor test

Experimental setup

SETTINGS:

Device:

DIE: D4

Arco: 7

Front-end circuit:

R_f : 500 k Ω

R_1 : 100 Ω

R_2 : 1 k Ω

PLCC68:

PIN 18, 27 $\longrightarrow V_{DC}$

PIN 24 $\longrightarrow GND$

PIN 22 $\longrightarrow v_{in}(t)$

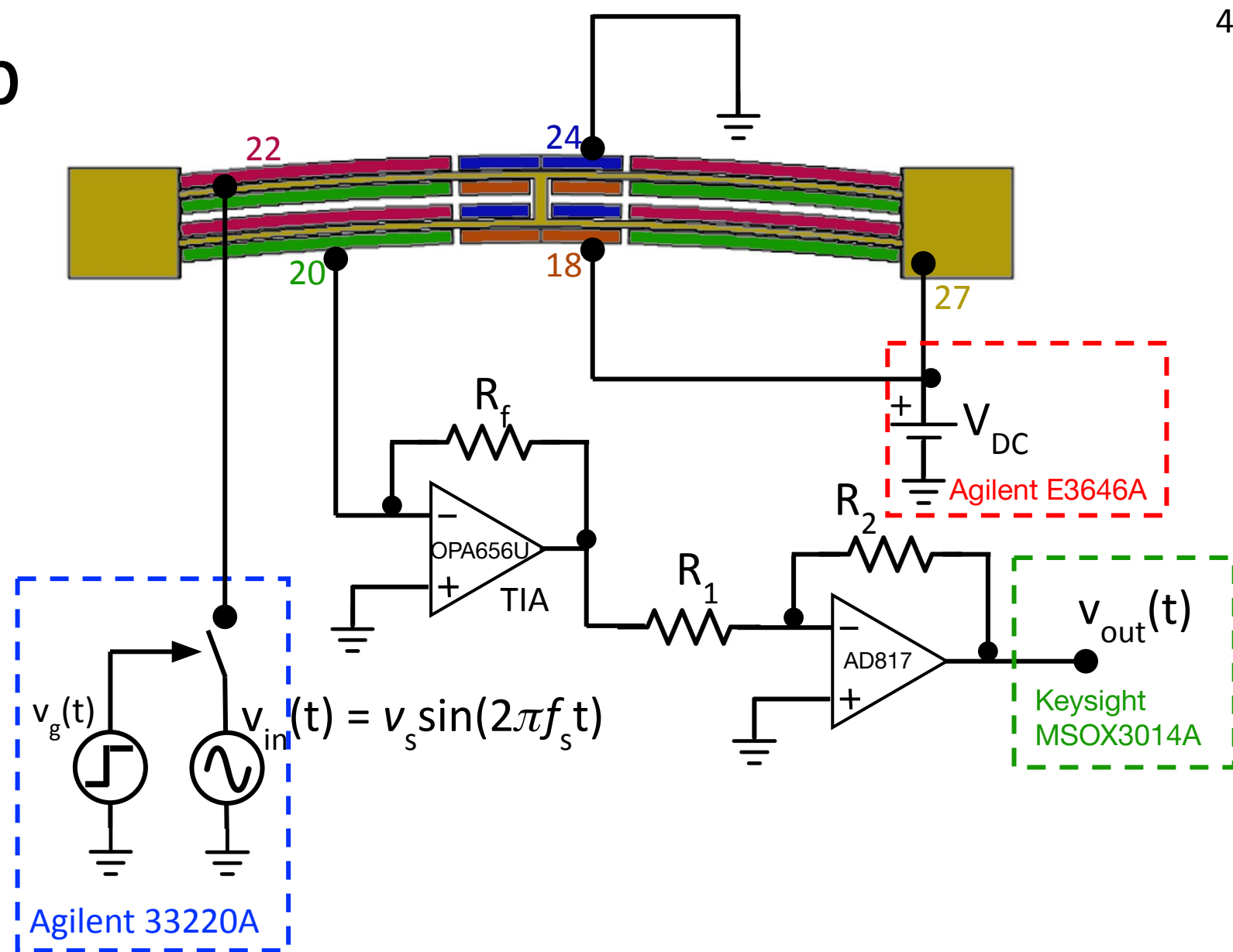
PIN 20 $\longrightarrow TIA$

Instrumentation:

Signal generator: Agilent 33220A

Power supply: Agilent E3646A

Oscilloscope: Keysight MSOX3014A



Operative conditions

5

Performed measurements:

Bias voltage V_{DC} : *start 3 V, stop 7 V, step 0.25 V*

Peak-peak amplitude v_s : *start 50 mV, stop 250 mV, step 10 mV*

$f_s = 418 \text{ kHz}$

Gate signal $v_g(t)$, Burst mode:

Amplitude: 0 - 3.3V,

Duty cycle: 50%

Period $T_g = 12 \text{ ms}$

Phase: 100 deg

Signal generator: $Z_{out} = 1 \text{ MOhm}$

Oscilloscope: $Z_{in} = 1 \text{ MOhm}$

Number of repeated measurements: $N = 500$

Exported csv file:

File name example:

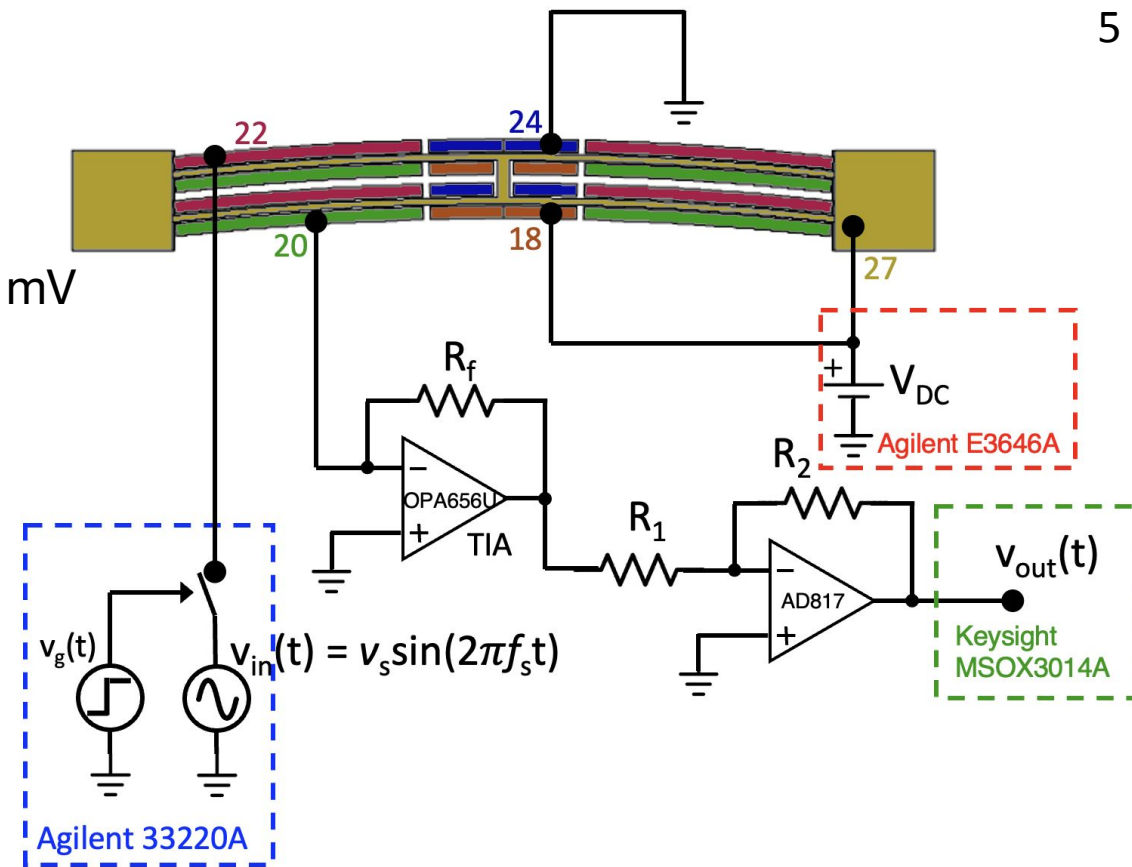
27_01_2026_OSC_50mV_Vdc_3_00V_Freq_Vt_0V_Freq_418000_Hz_Ph_100_deg_N500_T_12ms_Zsign_1MOhm_ZOscill_1MOhm_D4_Rf_500kOhm_G_10

Data format:

First column: gate signal v_g [V]

Second column: output signal v_{out} [V]

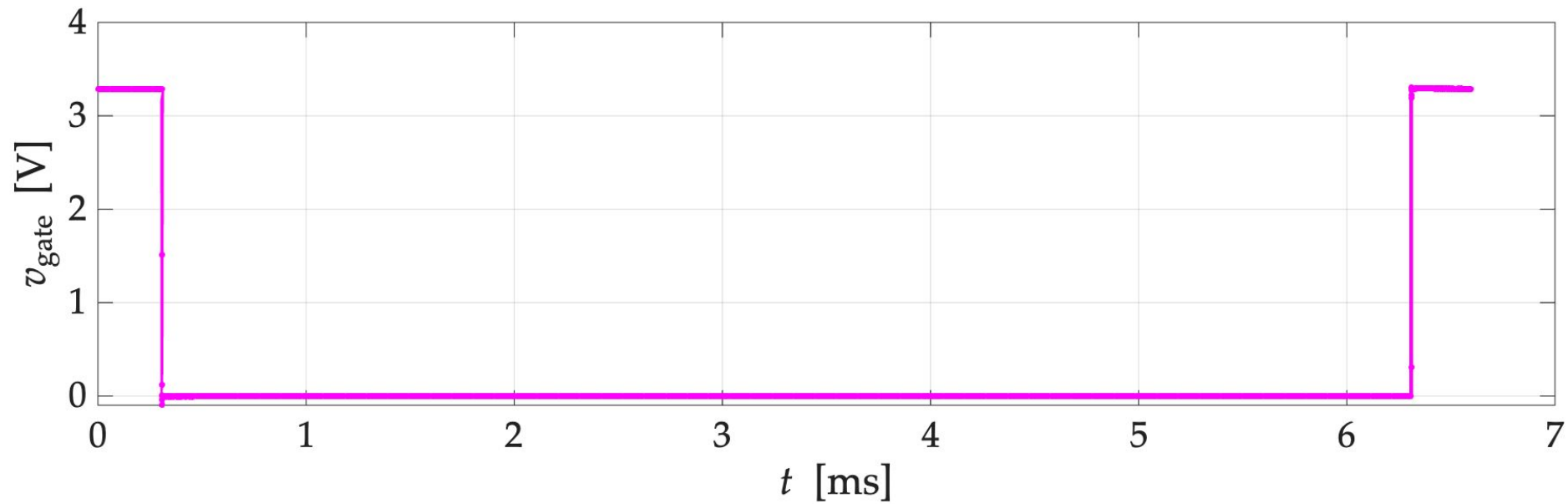
Third column: time [s]



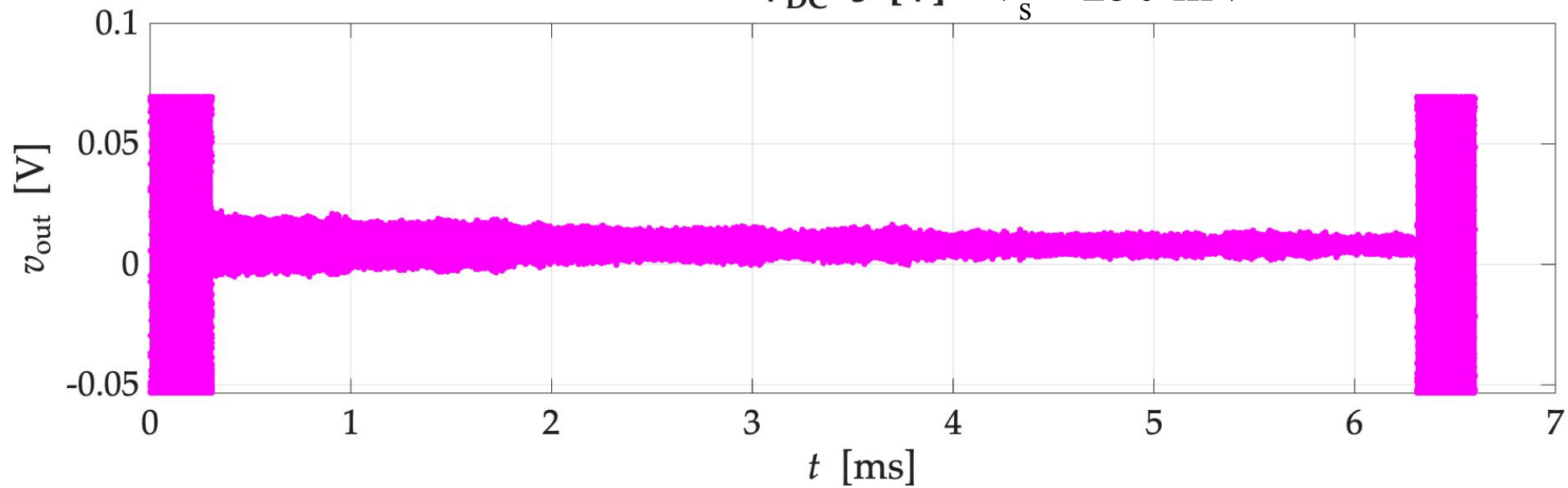
SHARED FOLDER:

27_01_2026_Time_Domain_Sweep_Vdc

Experimental results

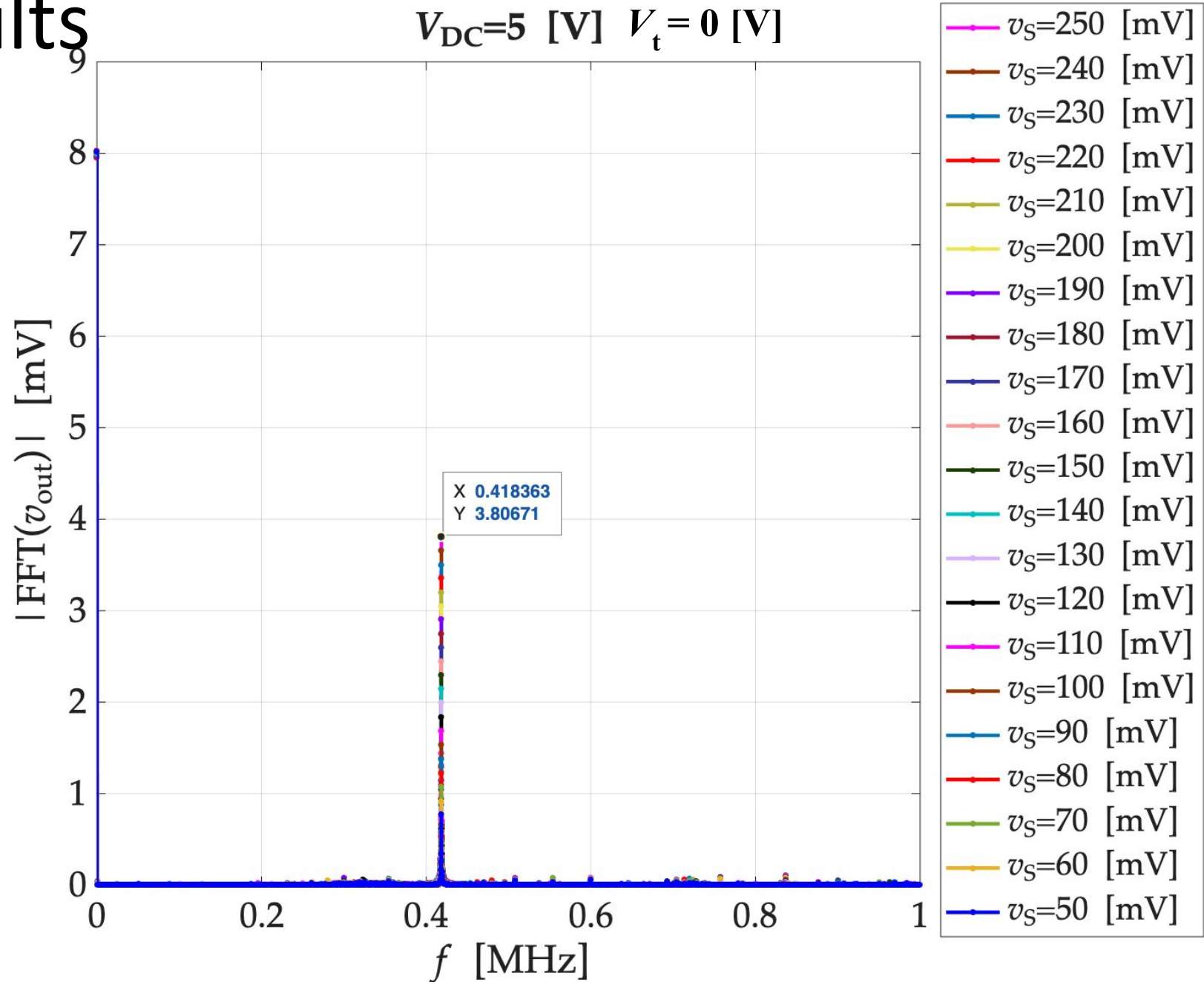


$V_{\text{DC}} = 5 \text{ [V]}$ $v_s = 250 \text{ mV}$



Experimental results

Unfiltered data

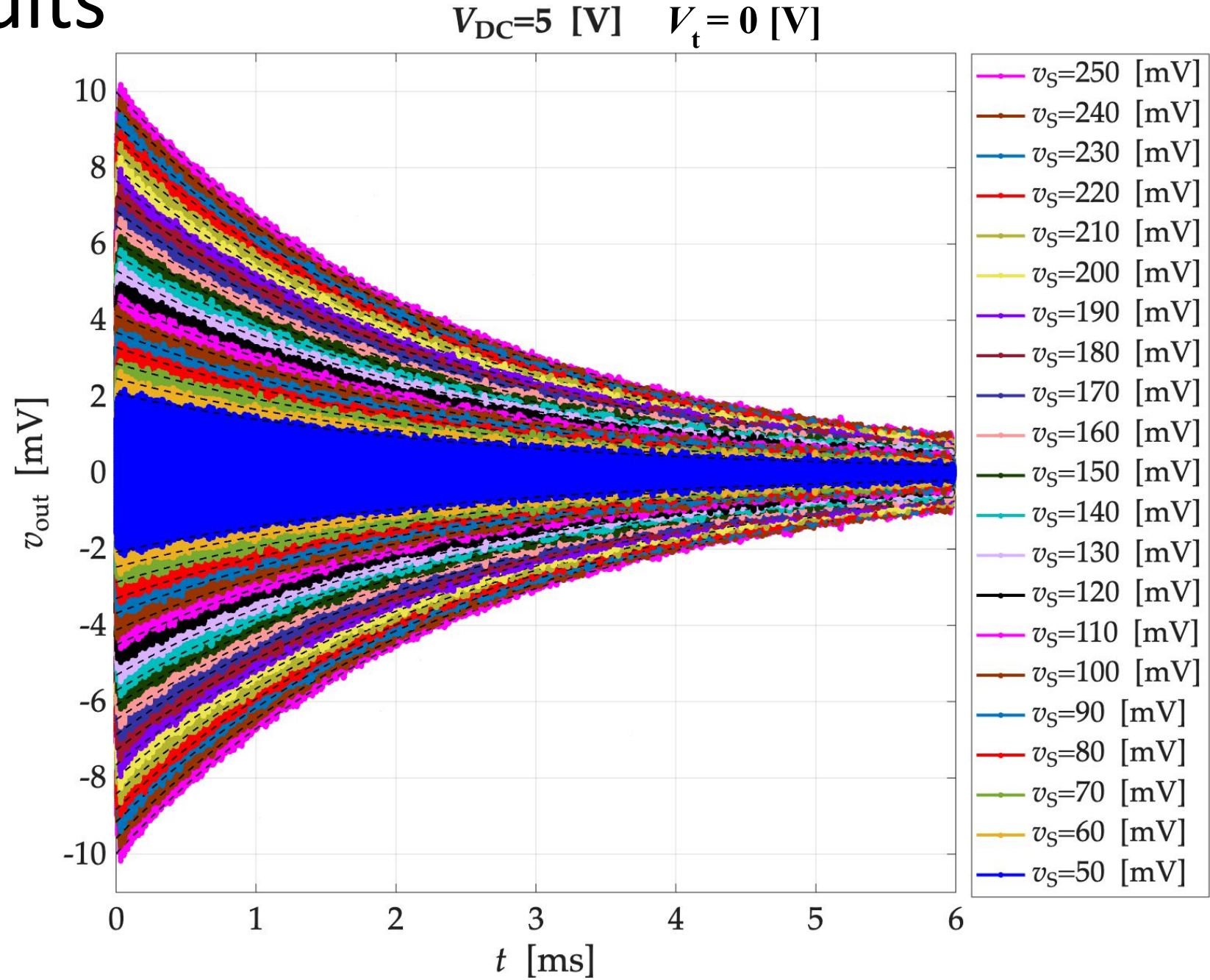


Experimental results

Bandpass IIR filter:

order: 18

BW = [378, 458] kHz

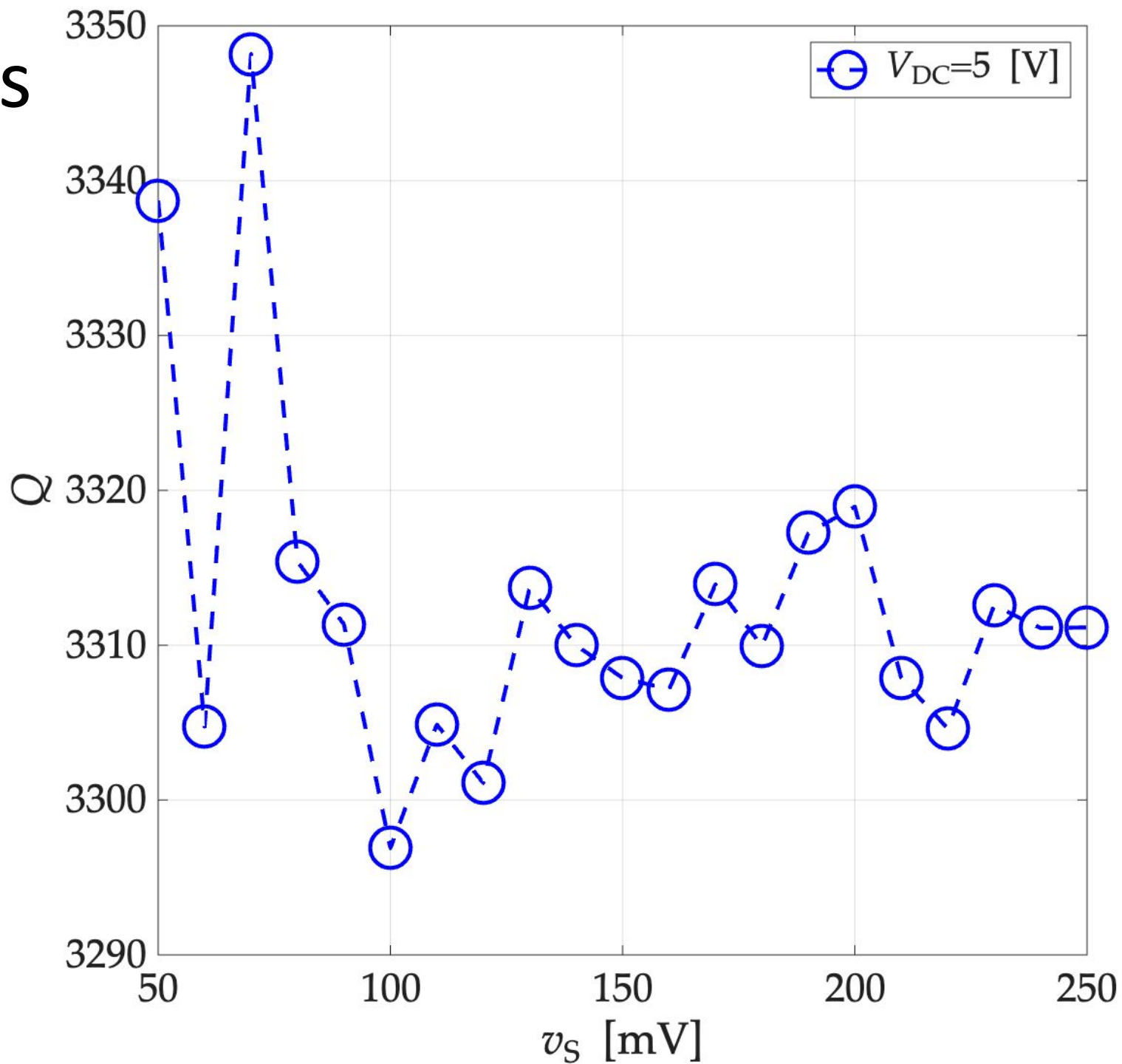


Experimental results

Bandpass IIR filter:

order: 18

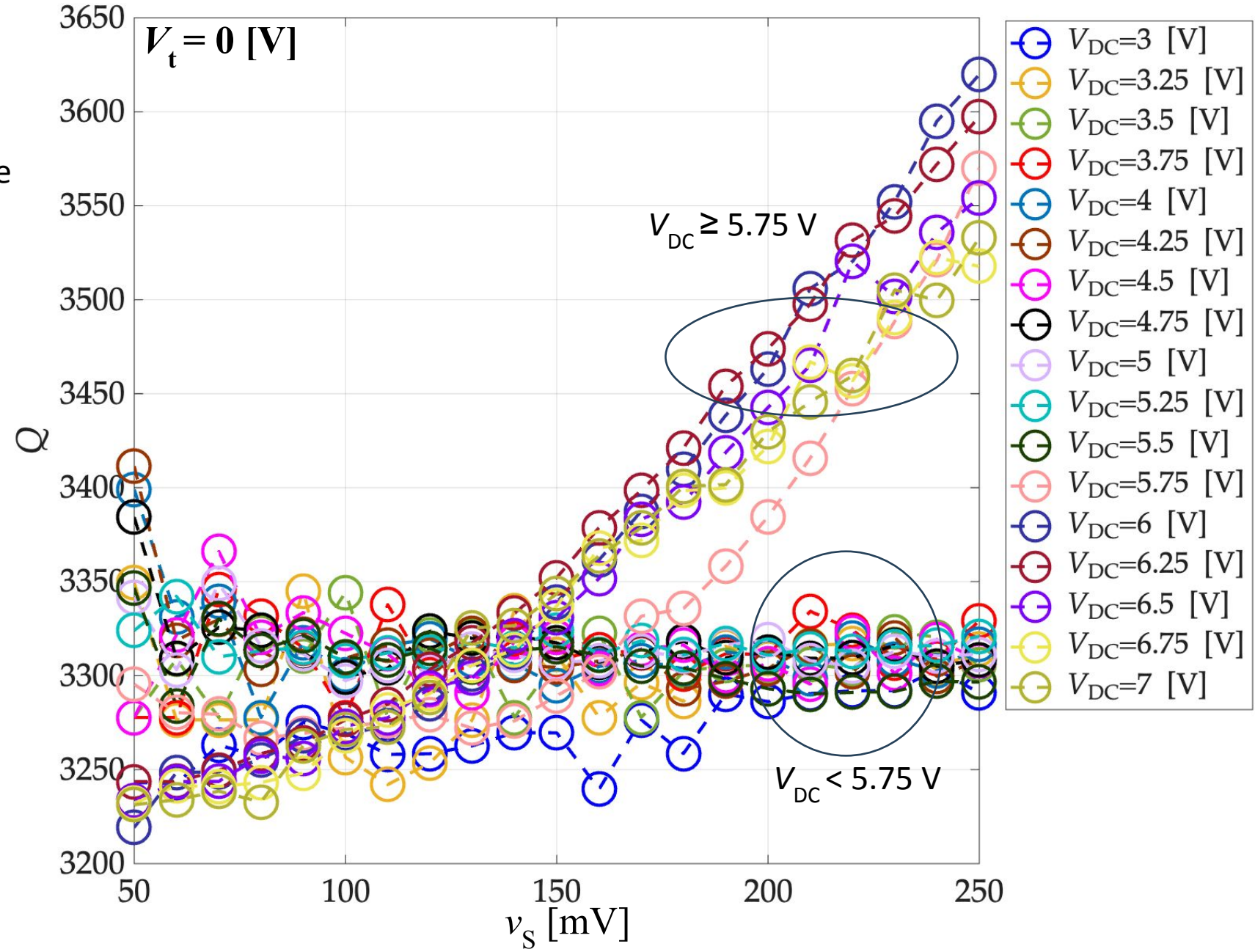
BW = [378, 458] kHz



Experimental results

NOTE:

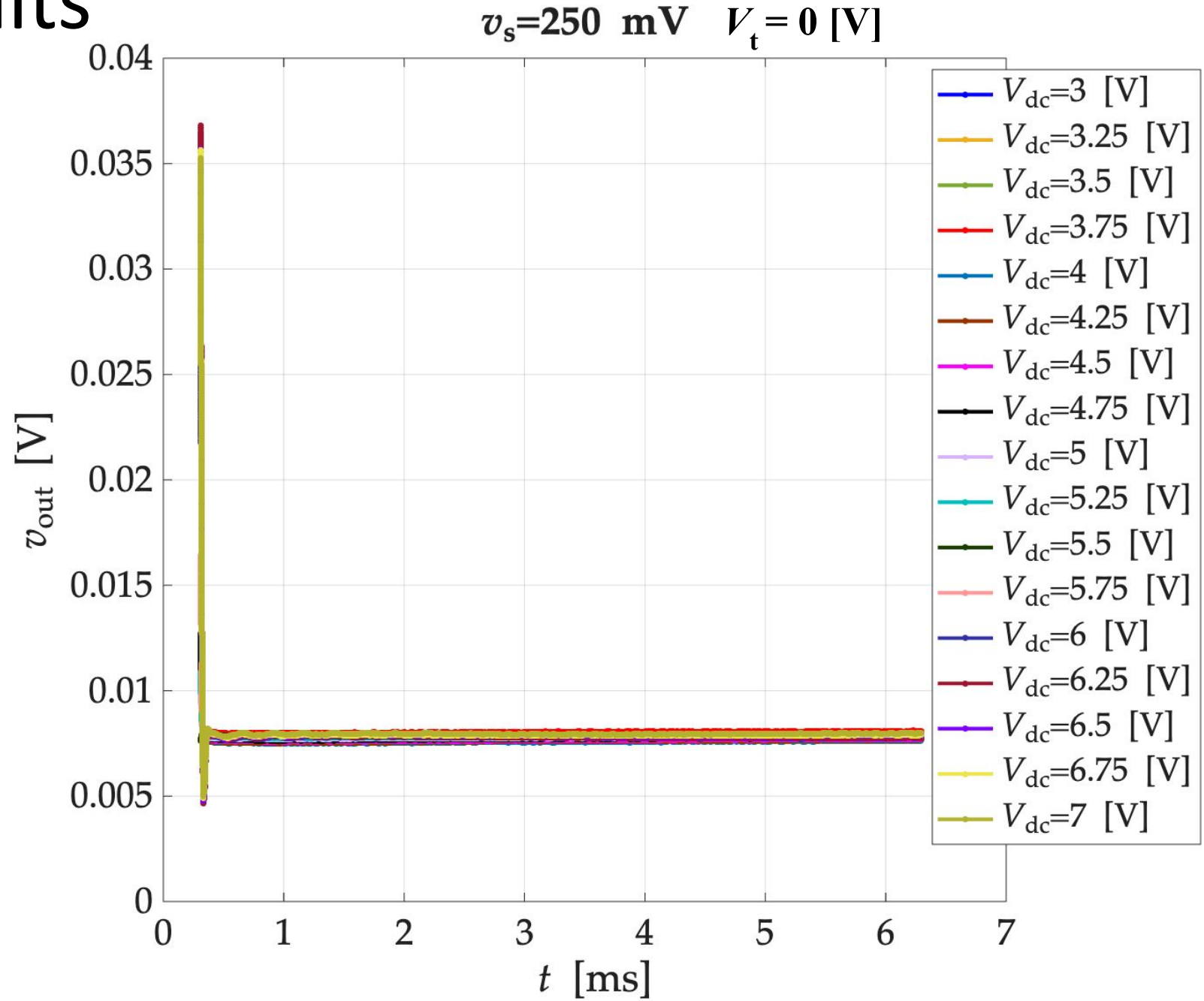
- For $V_{DC} < 5.75$ V Q values seems to be not function of v_s .
- For $V_{DC} \geq 5.75$ V Q values start to increase as v_s increases.



Experimental results

Offset investigation

Lowpass IIR filter only:
order: 2
BW = 0 - 10 kHz



1:2 resonance test – Frequency domain

Operative conditions

Performed measurements:

Bias voltage V_{DC} : 20.6 V

Tuning voltage V_T : 18 V

Peak amplitude v_s : *start* 850 mV, *stop* 1000 mV, *step* 50 mV

Sweep UP:

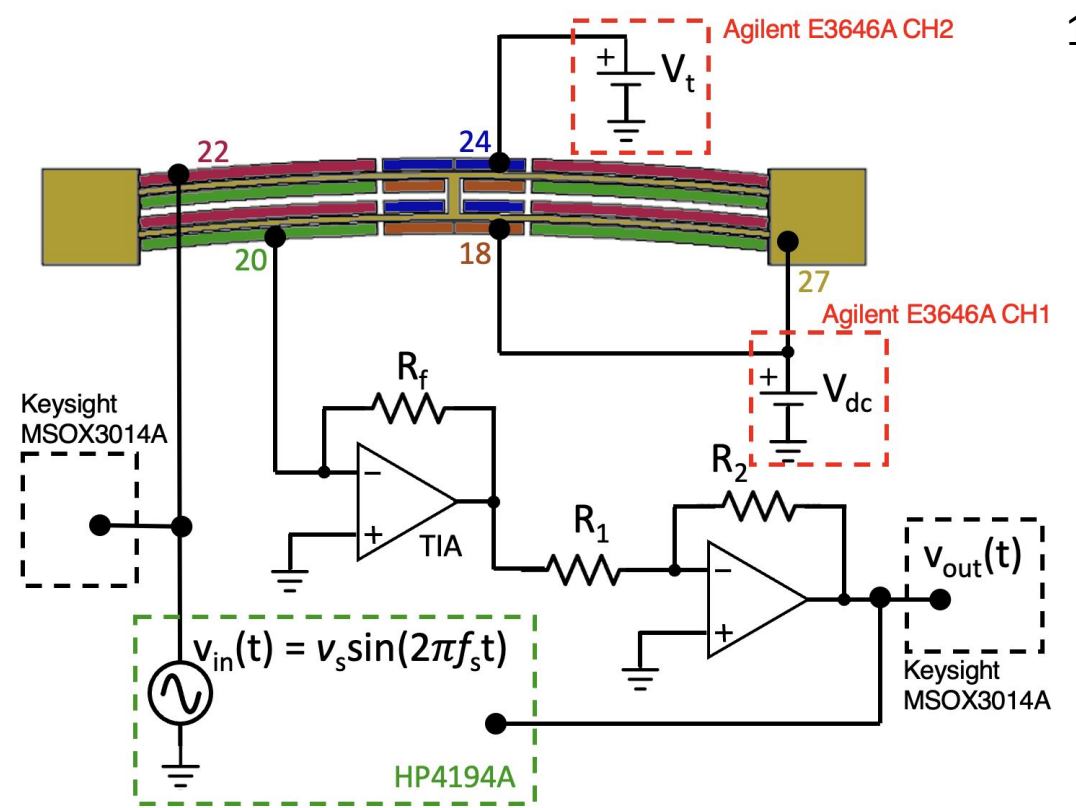
$f_s = \text{start } 414 \text{ kHz, stop } 422 \text{ kHz, step } 2 \text{ Hz}$

Sweep DOWN:

$f_s = \text{start } 407 \text{ kHz, stop } 422 \text{ kHz, step } 2 \text{ Hz}$

Oscilloscope: $Z_{in} = 50 \text{ Ohm}$

Number of repeated measurements: $N = 2$



Exported csv file:

File name example:

0_27_02_2026_Tch_20dB_1Mohm_Rch_20dB_1Mohm_D4_Rf_100kOhm_G_10_Osc_50Ohm_CH1_CH2_YT_Step_2Hz_Vdc_20.60V_Vt_18.00V_OSC_850.00mV_Short_AVG_2_Sweep_Up.csv

Data format:

1st column: magnitude of the transfer function $H(f) = V_{out}(f)/V_{in}(f)$

2nd column: standard deviation of the magnitude of the transfer function $H(f) = V_{out}(f)/V_{in}(f)$

3rd column: phase of the transfer function $H(f) = V_{out}(f)/V_{in}(f)$

4th column: standard deviation of the phase of the transfer function $H(f) = V_{out}(f)/V_{in}(f)$

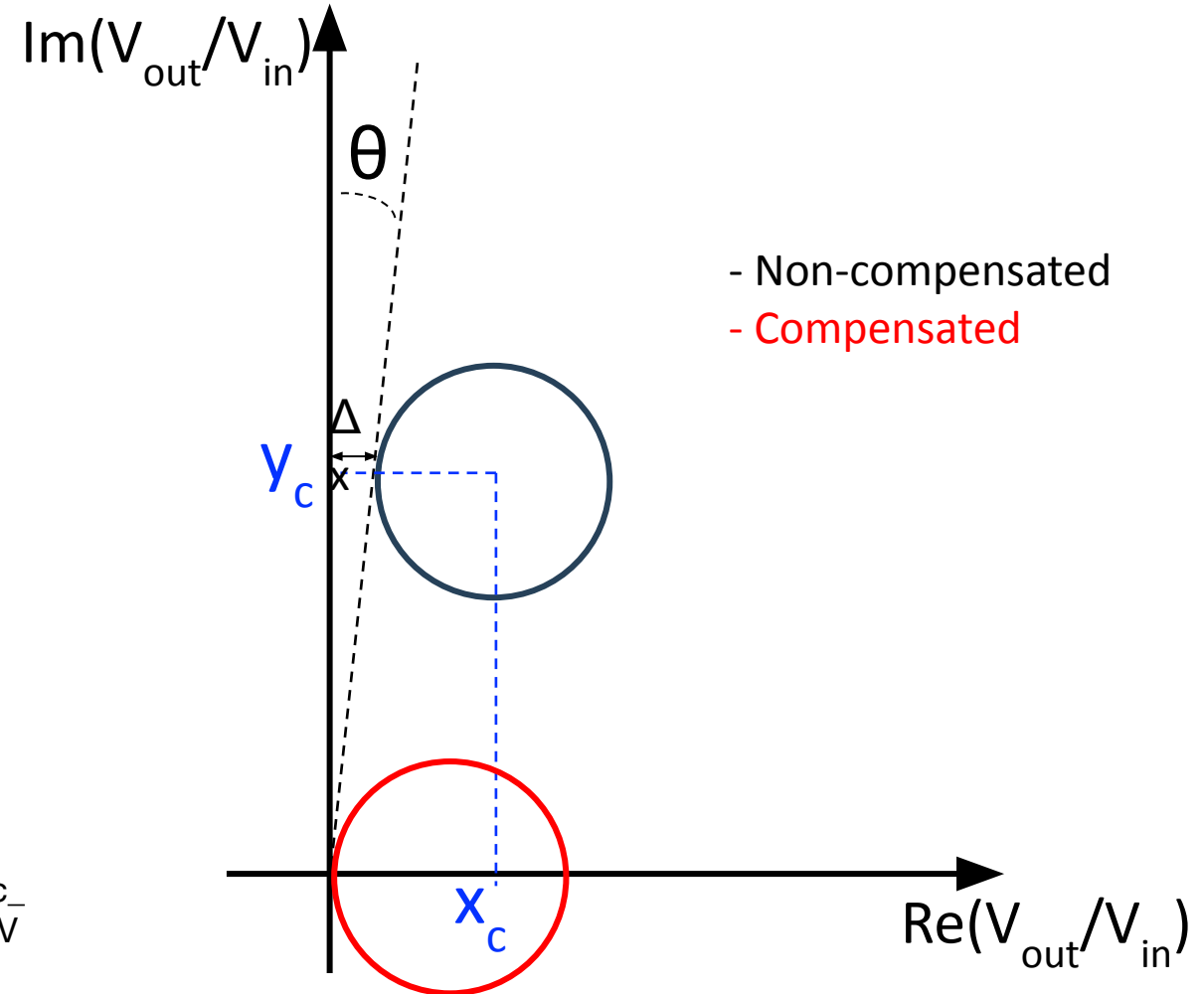
5th column: frequency of the sensing signal $v_{in}(t)$

SHARED FOLDER:

2026_02_27_Freq_1_2_Sweep_OSC

Numerical compensation procedure

1. Correction of the phase θ
2. Subtraction of the offset of the imaginary part y_c
3. Subtraction of the offset of the real part Δx



Exported csv file:

File name example:

0_27_02_2026_Tch_20dB_1Mohm_Rch_20dB_1Mohm_D4_Rf_100kOhm_G_10_Osc_50Ohm_CH1_CH2_YT_Step_2Hz_Vdc_20.60V_Vt_18.00V_OSC_850.00mV_Short_AV_G_2_Sweep_Up_Compensated.csv

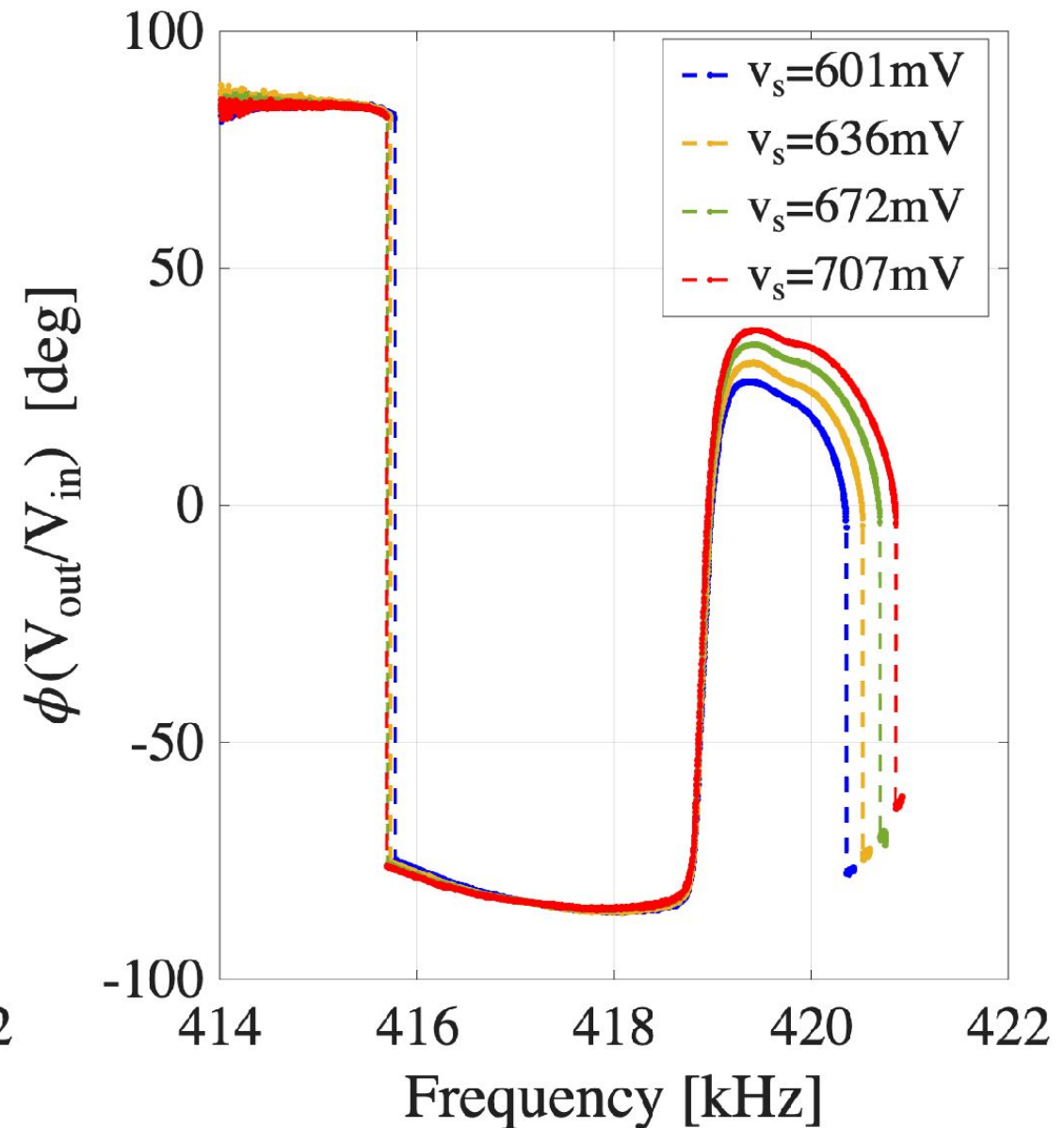
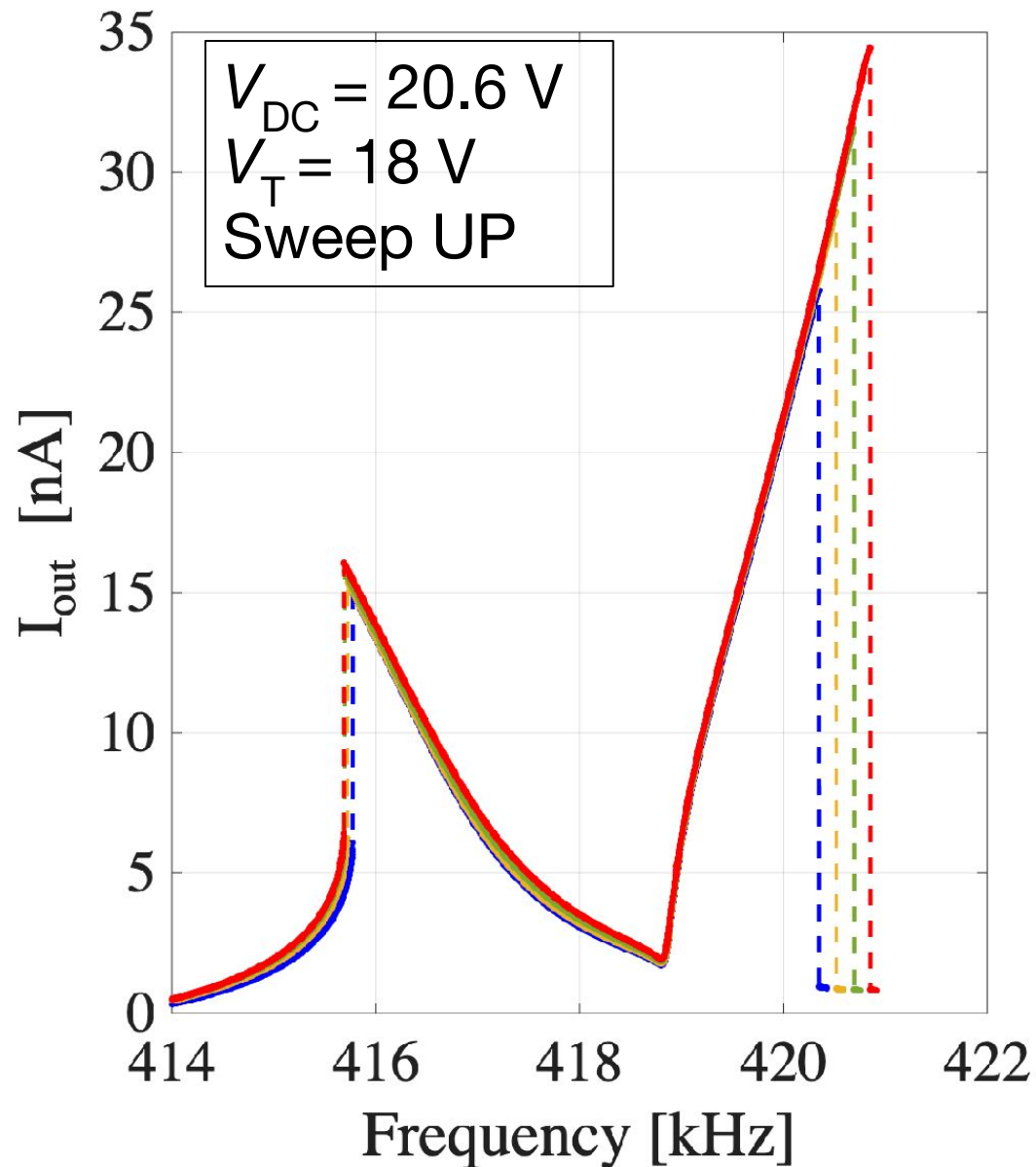
Data format:

1st column: rms value of the current $I_{\text{out}}(f)$

2nd column: phase of the transfer function $H(f) = V_{\text{out}}(f)/V_{\text{in}}(f)$

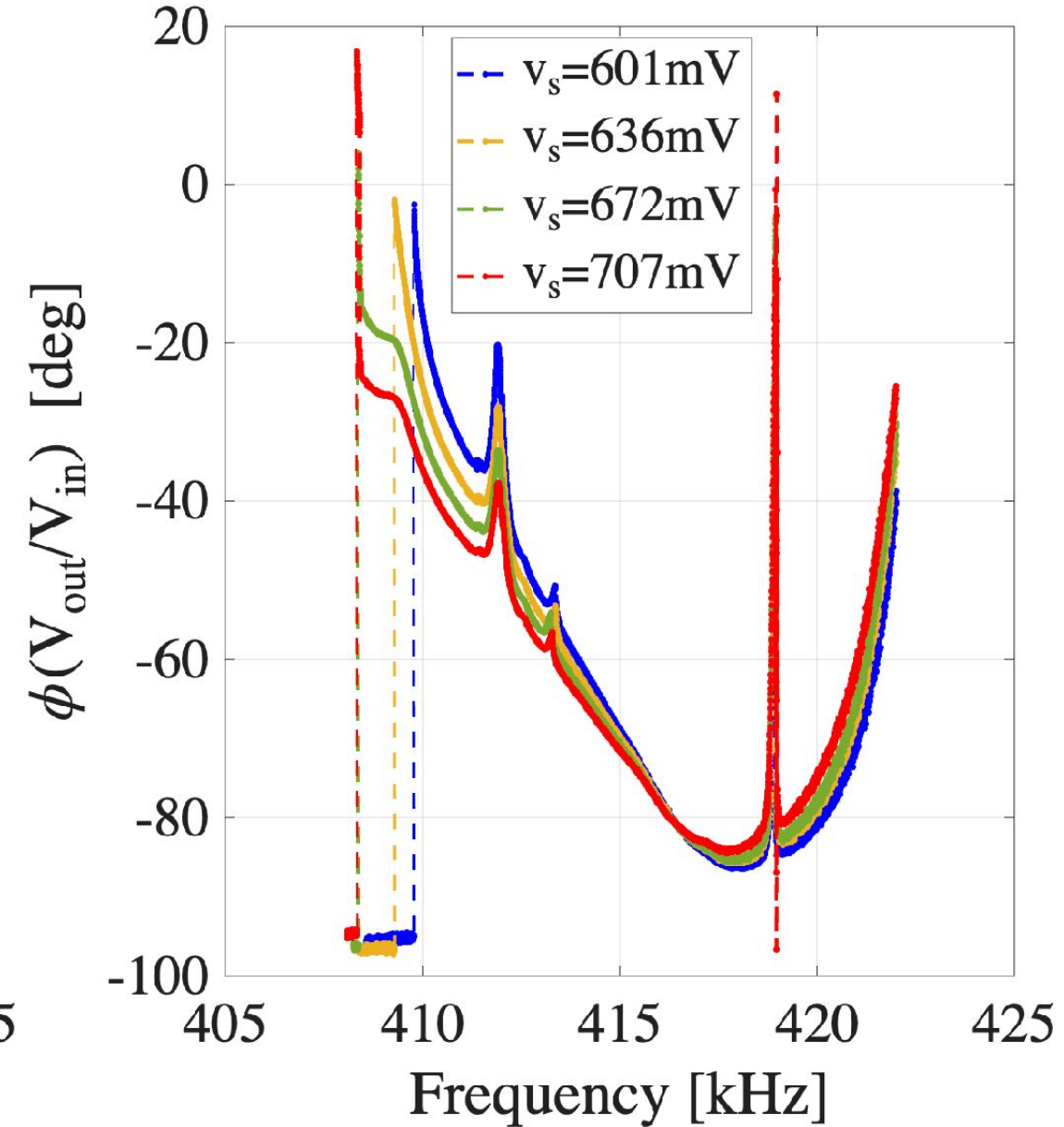
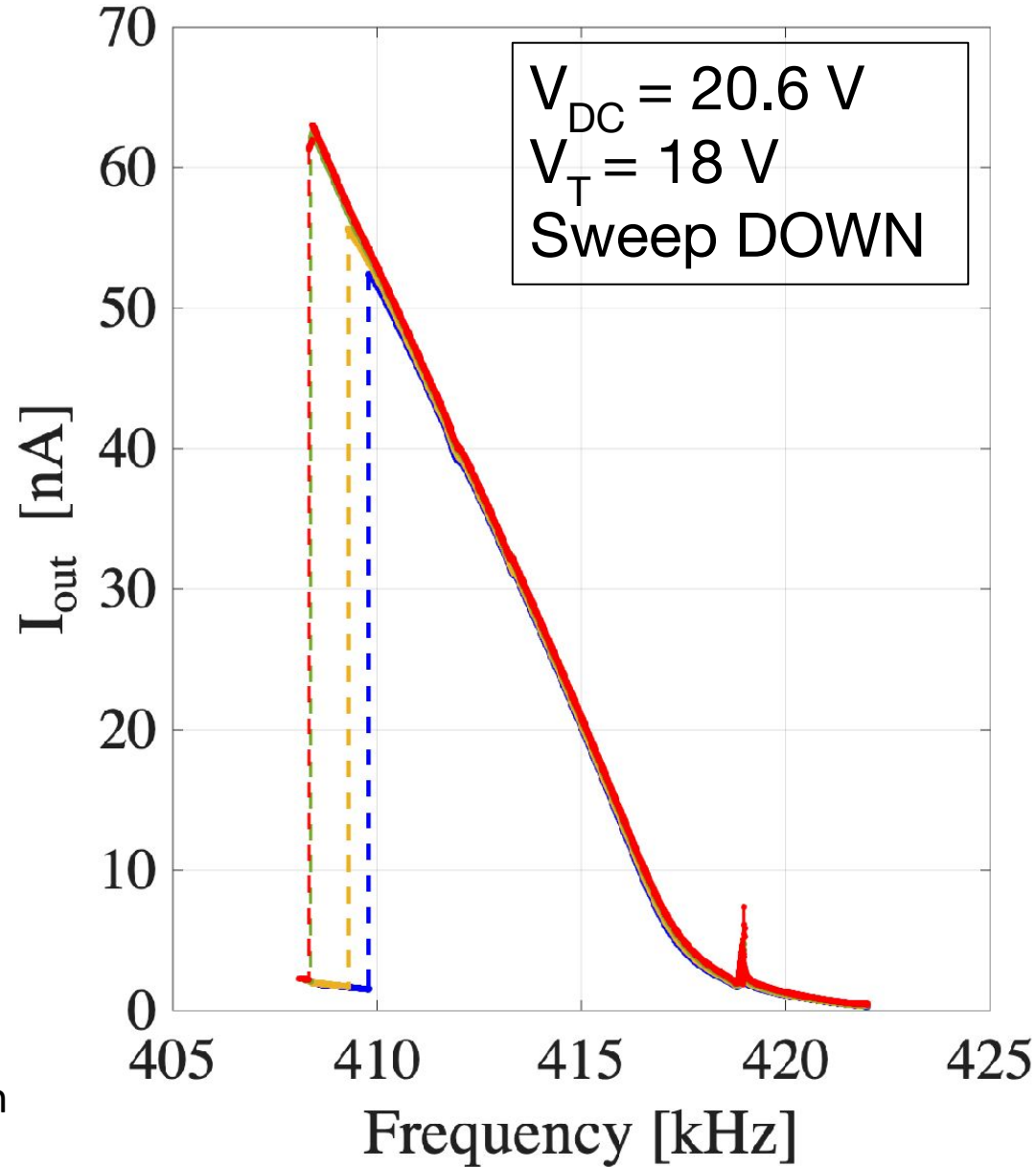
3rd column: frequency of the sensing signal $v_{\text{in}}(t)$

Example of numerically compensated measurements



NOTE:
 The rms
 amplitude of v_s
 has been shown
 in these plot

Example of numerically compensated measurements



NOTE:
 The rms
 amplitude of v_s
 has been shown
 in these plot

1:2 resonance test – Time domain

Experimental setup

SETTINGS:

Device:

DIE: D4

Arco: 7

Front-end circuit:

R_f : 100 k Ω

R_1 : 100 Ω

R_2 : 1 k Ω

PLCC68:

PIN 18, 27 $\longrightarrow V_{DC}$

PIN 24 $\longrightarrow V_t$

PIN 22 $\longrightarrow v_{in}(t)$

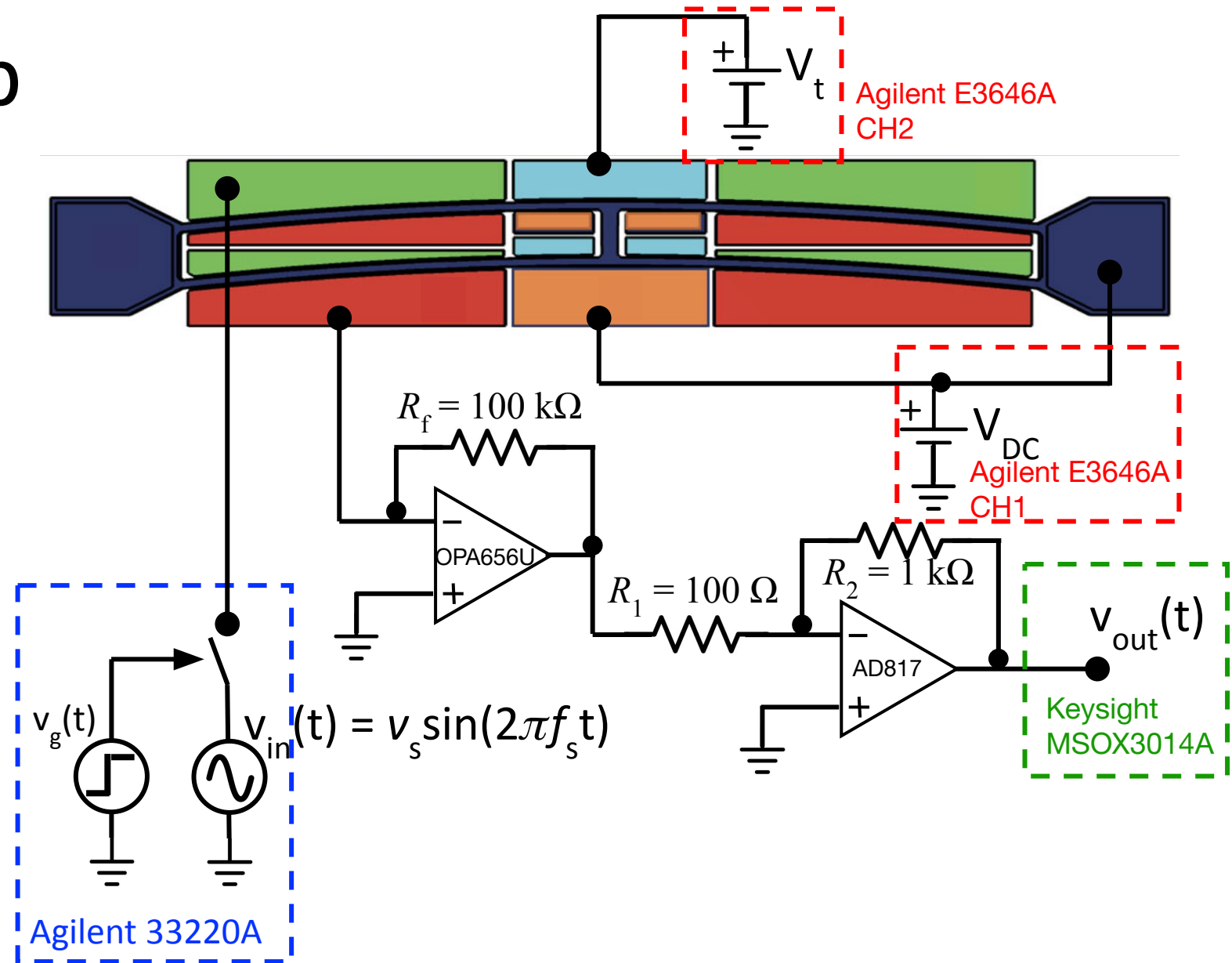
PIN 20 $\longrightarrow$ TIA

Instrumentation:

Signal generator: Agilent 33220A

Power supply: Agilent E3646A

Oscilloscope: Keysight MSOX3014A



Operative conditions

18

Performed measurements:

Bias voltage $V_{DC} = 20.6 \text{ V}$

Tuning voltage $V_t = 18 \text{ V}$

rms amplitude $v_s = 566 \text{ mV}$

f_s : start 418 kHz, stop 421 kHz, step 500 Hz

Gate signal $v_g(t)$, Burst mode:

Amplitude: 0 - 3.3V,

Duty cycle: 50%

Period $T_g = 12 \text{ ms}$

Phase: 100 deg @ $f_s = 418 \text{ kHz}$

Signal generator: $Z_{out} = 1 \text{ MOhm}$

Oscilloscope: $Z_{in} = 1 \text{ MOhm}$

Number of repeated measurements: $N = 500$

Exported csv file:

File name example:

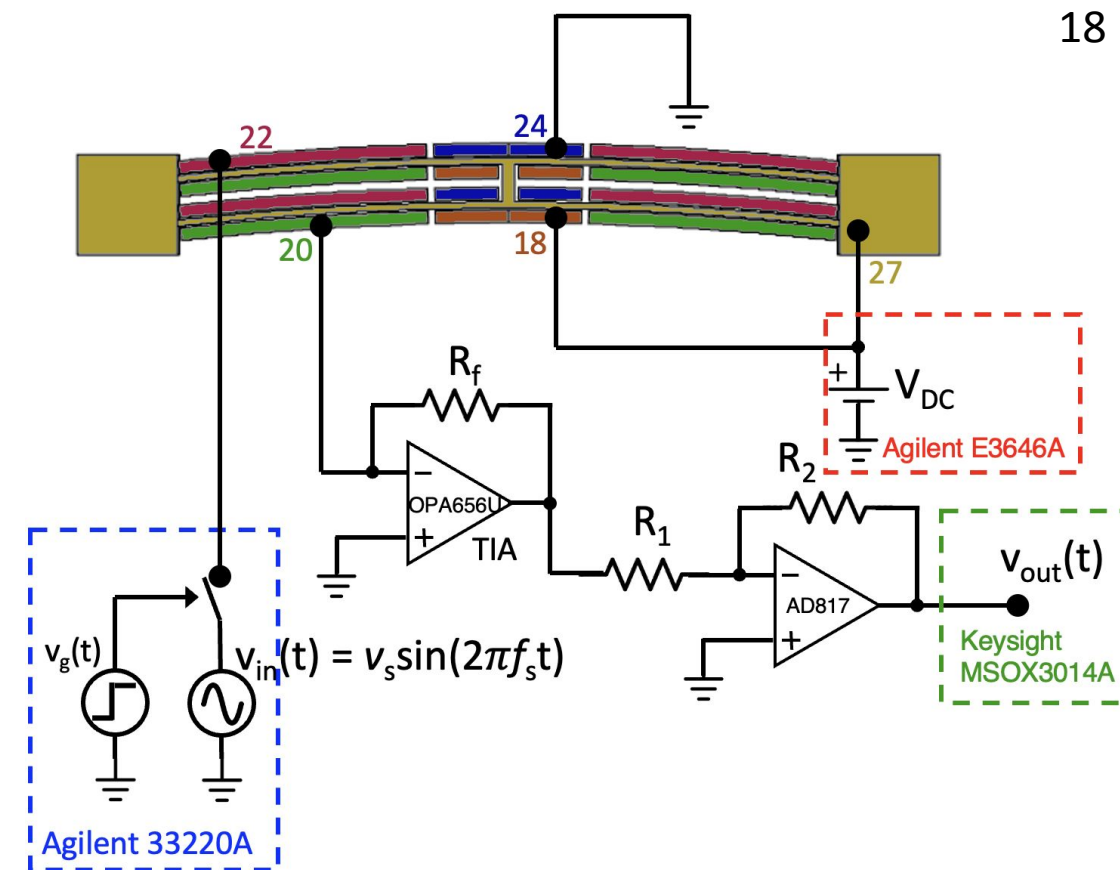
25_02_2026_OSC_1600mV_Vdc_20_60V_Vt_18V_Freq_418000_Hz_Ph_100_deg_N500_T_12ms_Zsign_1MOhm_ZOscill_1MOhm_D4_Rf_100kOhm_G_10

Data format:

First column: gate signal v_g [V]

Second column: output signal v_{out} [V]

Third column: time [s]

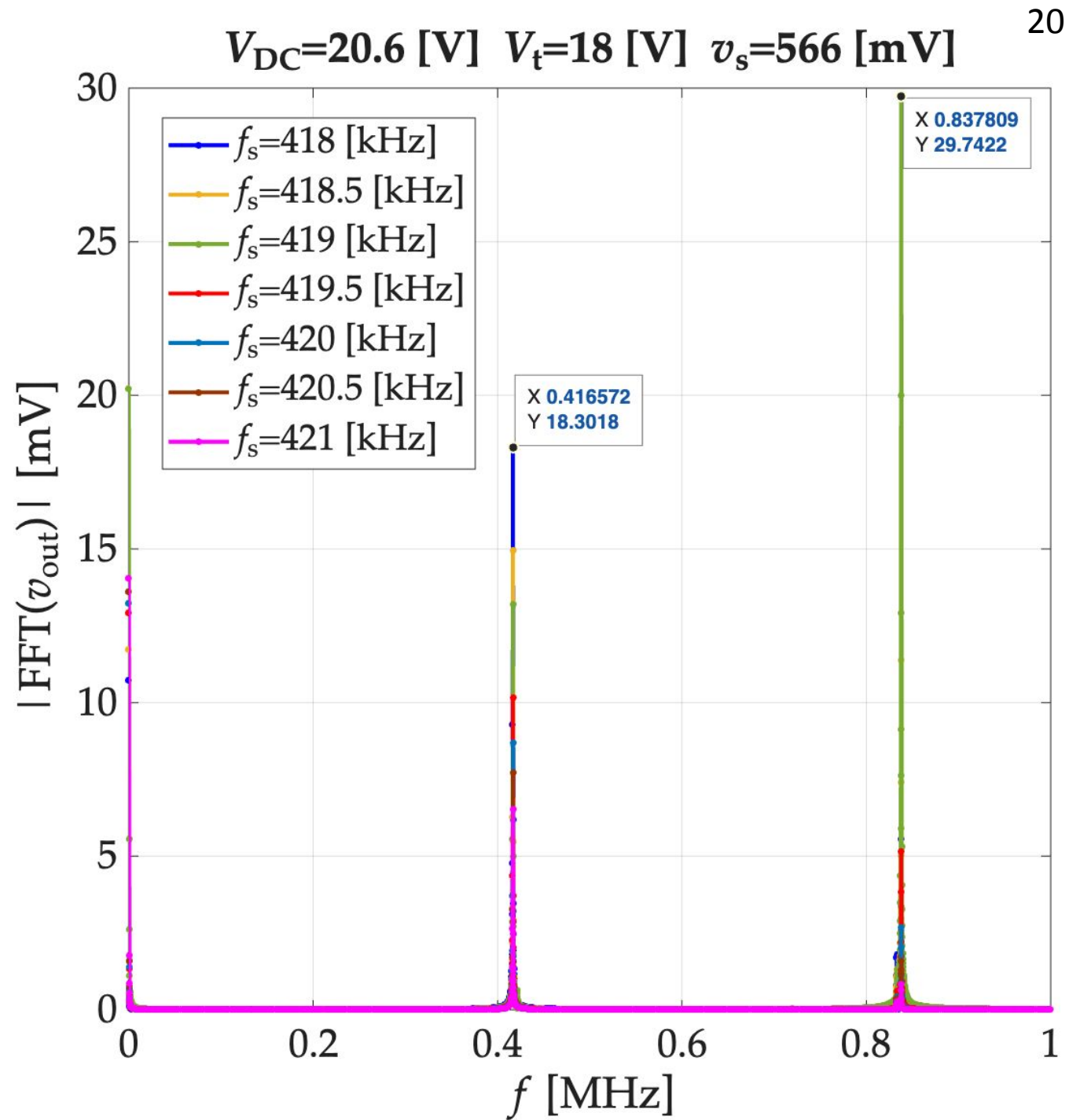


SHARED FOLDER:

2026_03_03_Time_Sweep_Freq_1_2

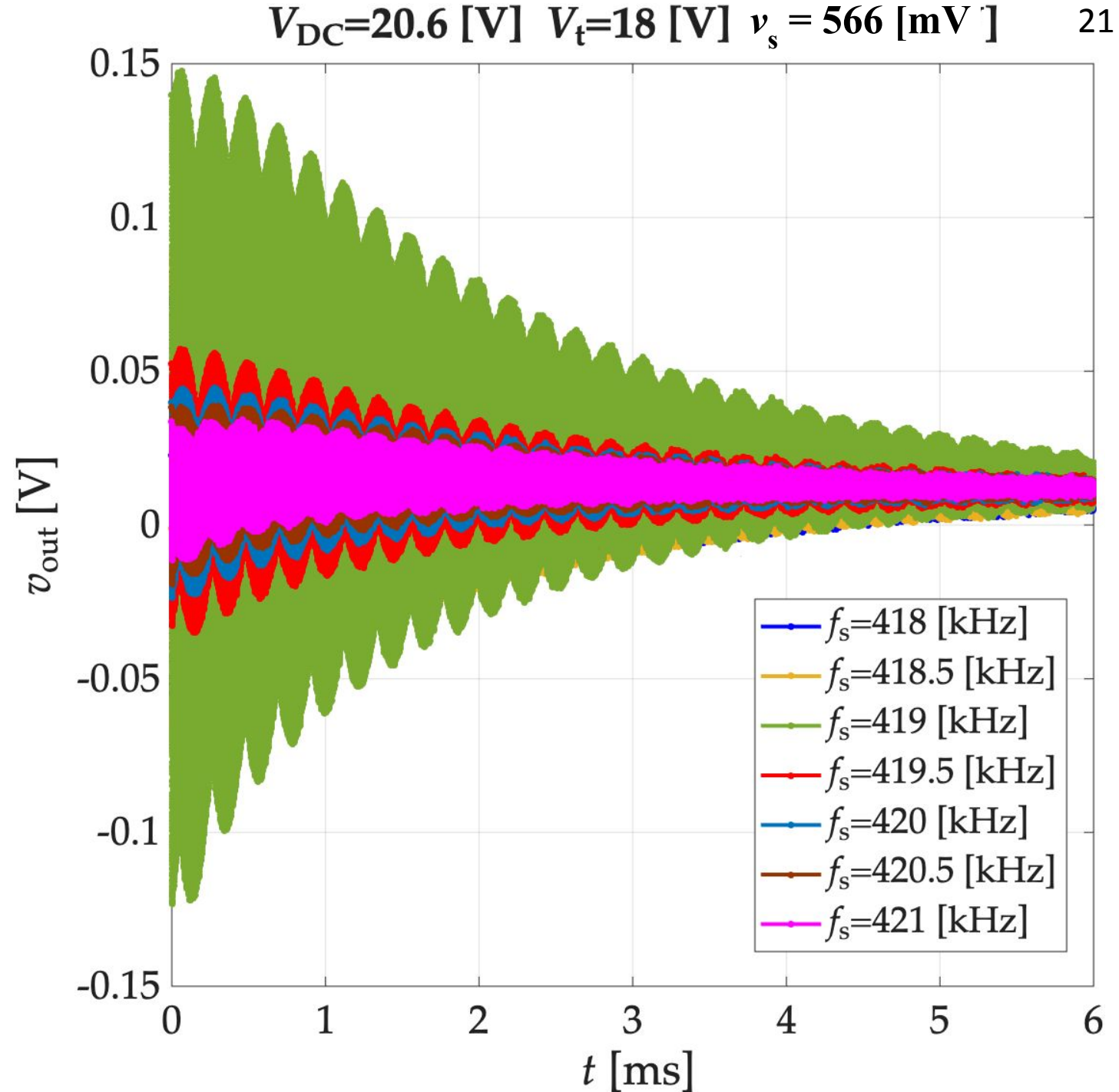
Experimental results

Unfiltered data



Experimental results

Unfiltered data



Experimental results

Offset investigation

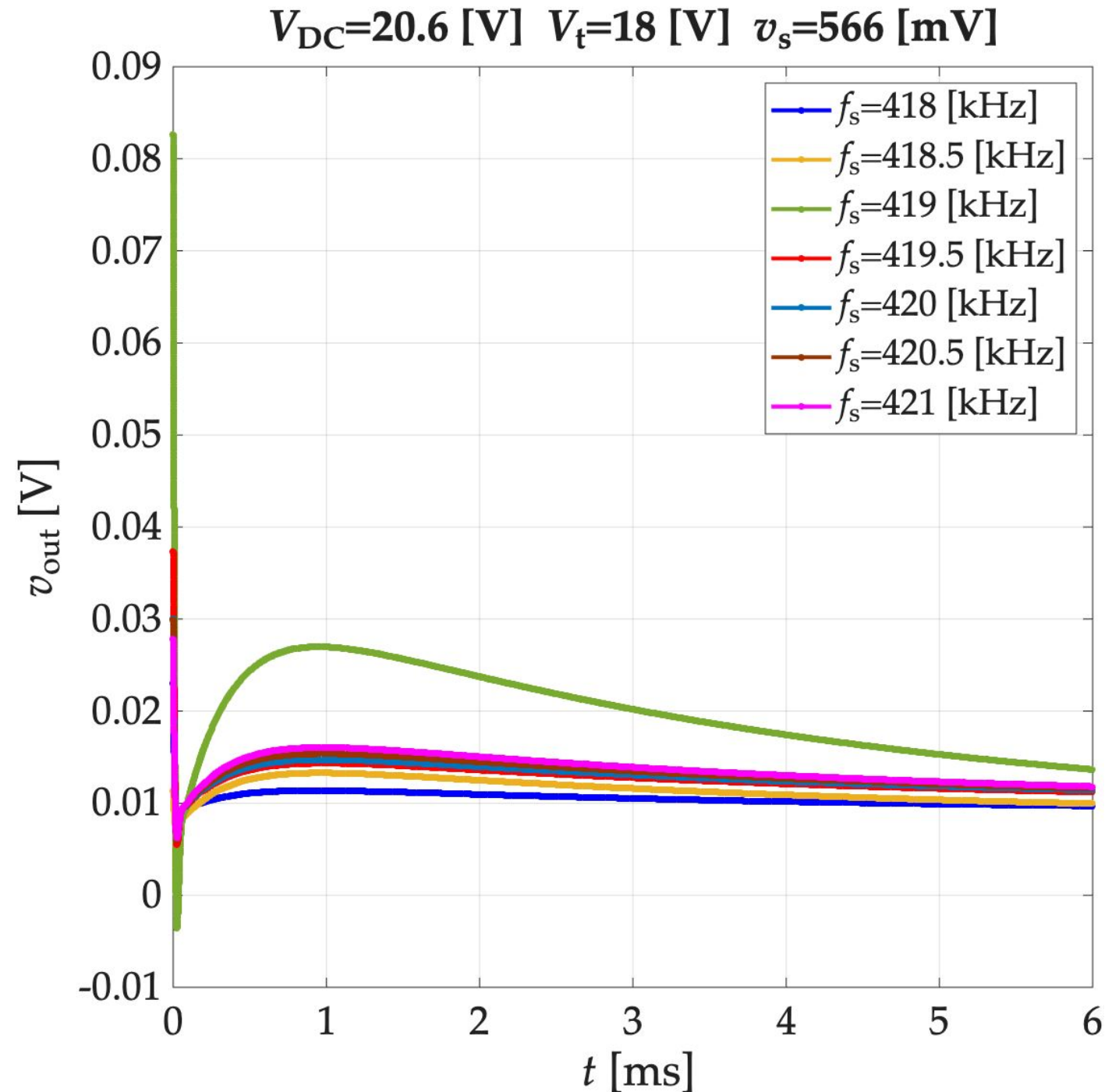
Lowpass IIR filter only:

order: 2

BW = 0 - 10 kHz

The offset behavior is probably due to the coupling with the impedance oscilloscope.

It can be removed with a filter



Experimental setup

